University of Leeds
School of Electronic and Electrical Engineering

ELEC5870M Interim Report

Fabrication of Enhancement Mode HEMT Devices via Gate Recession

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Abstract

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Acknowledgements

Abbreviations

2DEG	Two-dimensional electron gas
HEMT	High-electron-mobility transistor

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CHAPTER 1

Introduction

Modern communication, sensing, and microwave systems impose increasingly demanding requirements on the semiconductor devices that form their front-end signal paths. Higher operating frequencies, tighter noise constraints, and strict linearity targets continue to push transistor technologies beyond the capabilities of many silicon-based platforms. [1] Although silicon processes have made notable progress in recent years, they often fall short in providing the necessary electron mobility, noise performance, or high-frequency gain required for performance-critical microwave and millimetre-wave applications. (e.g. silicon's 1200 cm^2/Vs compared to GaAs's 6500 cm^2/Vs in transistors for MMICs) [1]

III–V High–Electron–Mobility Transistors (HEMTs) play a central role in addressing these challenges. Their heterostructure–based design creates a highly conductive channel at the interface between two semiconductor layers with different bandgaps. At this interface a two–dimensional electron gas (2DEG) forms, supporting exceptionally high carrier mobility. The resulting strong transconductance, stable high–frequency behaviour and low–noise performance have firmly established HEMTs as core devices throughout microwave and millimetre–wave circuit design. [2]

Research activity within the field has increasingly shifted toward gallium nitride (GaN) HEMTs, driven by the material properties enabled by its wide bandgap. GaN supports high breakdown voltages, strong power–handling capability and robust operation under large electric fields, making it the preferred choice for many high–power and high–linearity applications [3]. Despite this growing dominance, gallium arsenide (GaAs) HEMTs remain one of the most mature and well characterised III–V transistor platforms. Their stability, extensive documentation and long industrial history continue to make them an invaluable system for studying the relationships between heterostructure design, device architecture and electrical behaviour. [4, 5]

For the purposes of this project, GaAs presents clear practical and technical advantages. The Leeds Nanotechnology Cleanroom maintains established, reliable processes for GaAs/AlGaAs heterostructures, and suitable wafers are readily available with reproducible epitaxial quality [6].

Standard GaAs/AlGaAs HEMTs are inherently depletion-mode devices; they naturally conduct current at zero gate bias and require a negative voltage to switch off. [2] This 'normally-on' behaviour is often undesirable as it necessitates complex negative polarity power supplies [7] and complicates the design of direct-coupled logic circuits. [8] Consequently, there is significant interest in developing enhancement-mode devices that are 'normally-off' and offer reduced power consumption.

One common approach to achieving enhancement-mode operation is to recess the gate into the AlGaAs donor layer, thereby reducing the amount of charge carriers available to form the channel and shifting the threshold voltage in the positive direction. [9] However, achieving enhancement-mode operation through gate recessing is inherently non-trivial, as it requires precise and repeatable removal of only a few nanometres of material, with small variations in etch depth leading to significant changes in threshold voltage and device behaviour. [10]

The primary objective of this project is to investigate the relationship between gate recess depth and threshold voltage in GaAs HEMTs, with a focus on understanding the practical limitations and repeatability of the fabrication process. This will be achieved through the development of a gate recession process in which the gate contact is recessed into the AlGaAs donor layer to shift the threshold voltage into the positive region.

To provide a rigorous comparison between depletion-mode and enhancement-mode devices, this interim report first focuses on the fabrication and characterisation of a series of non-recessed depletion-mode devices, achieved in first semester. (Chapters 3 and 4). These standard depletion-mode devices serve as a control to validate the HEMT fabrication process and establish reference parameters throughout this project. Finally, Chapter 5 outlines the theory and proposed process strategy required to implement the precise donor layer etch needed to achieve enhancement-mode operation in the second phase of this work.

CHAPTER 2

Technical Background

2.1 Overview of GaAs/AlGaAs HEMTs

In the GaAs/AlGaAs HEMTs used in this project, the interface between the GaAs channel layer and the overlying AlGaAs barrier produces a discontinuity in the conduction band because the two materials possess different bandgaps. This offset creates a potential well on the GaAs side of the junction, as illustrated in Figure 2.1, which highlights the conduction band discontinuity responsible for carrier confinement and 2DEG formation at the heterointerface. Electrons supplied by silicon donors in the modulation doped AlGaAs layer transfer into this well and accumulate to form a two dimensional electron gas (2DEG) that serves as the conducting channel. The formation of this spatially confined, high mobility 2DEG at the heterointerface is the defining feature of the HEMT structure [11].

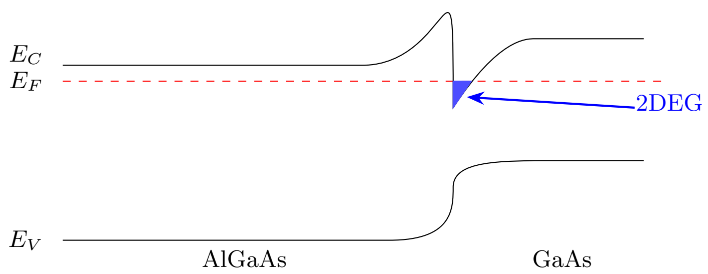

Figure 2.1: Band Structure Diagram of the AlGaAs/GaAs interface demonstrating the resulting Two-Dimensional Electron Gas (2DEG) formed in a Potential Well.

2.2 The GaAs/AlGaAs Heterostructure

The HEMTs investigated in this project are based on a conventional GaAs/AlGaAs heterostructure in which a sequence of epitaxial layers is engineered to create and control a 2DEG. A schematic cross-section of the structure is shown in Figure 2.2, highlighting the cap, donor, spacer and channel layers, together with their relative thicknesses, that determine the electrostatic control and transport behaviour of the device.

At the surface of the structure lies a 10 nm thick, highly doped GaAs cap layer. This cap provides a stable and chemically robust surface for forming the Schottky gate contact, whilst also preventing oxidation of the underlying AlGaAs. Its high conductivity reduces series resistance in the access regions adjacent to the source and drain contacts, thereby improving current injection into the channel.

Beneath the cap layer is a 40 nm thick, silicon doped AlGaAs donor layer. Silicon donors

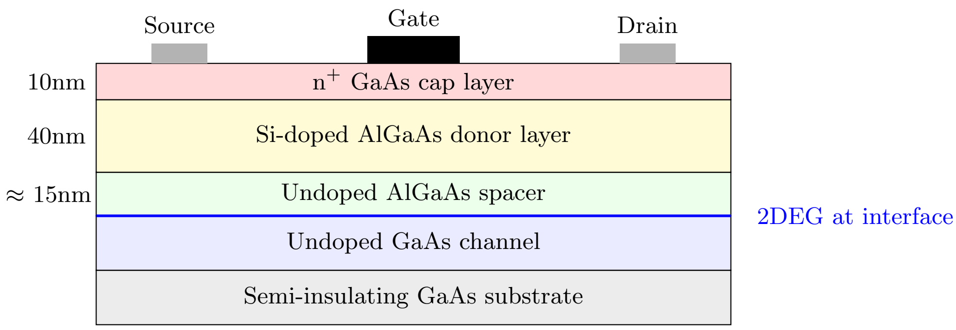

Figure 2.2: Cross-section of the GaAs/AlGaAs HEMT used in this project (not to scale).

in AlGaAs act as electron suppliers, with carriers transferring into the GaAs channel as a result of the conduction band discontinuity at the heterointerface. This approach is known as modulation doping, as the dopants are spatially separated from the channel rather than placed directly within it. Both the donor concentration and its separation from the interface strongly influence the resulting sheet carrier density of the 2DEG, and therefore the threshold voltage and maximum drain current of the device.

A 15 nm thick undoped AlGaAs spacer layer separates the donor layer from the GaAs channel. This layer is critical to device performance; by physically isolating the 2DEG from the ionised silicon dopants, it significantly suppresses Coulomb scattering [12]. This spatial separation allows carriers to propagate through the channel with exceptionally high mobility, which is a defining advantage of GaAs/AlGaAs HEMTs. The effect is particularly pronounced at cryogenic temperatures, where phonon scattering is minimised and impurity scattering would otherwise dominate transport behaviour [13]. The spacer thickness therefore represents a fundamental design trade off; increasing the spacer thickness improves mobility but weakens electrostatic coupling between the gate and the channel, reducing transconductance.

The GaAs channel beneath the spacer hosts the 2DEG itself. Electrons are confined at the AlGaAs/GaAs interface by the conduction band offset, forming a narrow quantum well in which transport occurs. The depth, density and confinement of this interfacial channel collectively govern the electrical behaviour of the device and form the physical basis for the threshold voltage extraction and transfer characteristics analysed later in this report.

2.3 HEMT Operating Principles

The electrical operation of GaAs/AlGaAs HEMTs is governed by the interaction between the Schottky gate contact and the 2DEG within the heterostructure. When the gate metal is deposited onto the GaAs surface, a Schottky barrier is formed, producing a depletion region beneath the gate. Application of a gate voltage modulates the width of this depletion region, enabling electrostatic control over the carrier density in the 2DEG at the AlGaAs/GaAs interface [14]. This gate induced modulation underpins the field effect behaviour of the device.

2.3.1 Depletion Mode Operation

The HEMTs fabricated during the first semester of this project operate in depletion mode. In the conventional heterostructure described above, the combined thickness of the GaAs cap and the AlGaAs donor layer, totalling approximately 50 nm, exceeds the zero bias depletion width of the Schottky gate. As a result, the 2DEG remains populated at zero gate bias, and the device naturally conducts when $V_G = 0$ [5].

To terminate conduction, a negative gate voltage must be applied to extend the depletion region until it physically intersects the GaAs channel. The specific gate voltage at which the carrier concentration in the channel is fully depleted defines the threshold voltage, V_{th}. In depletion mode devices, this threshold voltage lies in the negative bias region, a characteristic that directly motivates the transition towards enhancement mode operation explored in this project.

2.3.2 Enhancement Mode Operation

Enhancement mode operation is achieved when the device is normally off, requiring a positive gate voltage to initiate conduction. Fundamentally, the transition from depletion mode to enhancement mode corresponds to a reduction in the effective gate to channel separation distance, d. One widely adopted method for achieving this is gate recession, in which the GaAs cap layer and a controlled thickness of the AlGaAs donor layer are selectively etched beneath the gate region.

By reducing d, the built-in potential of the Schottky contact becomes sufficient to fully deplete the 2DEG at zero gate bias, shifting the threshold voltage into the positive region [9]. Under these conditions, conduction occurs only when a positive gate voltage is applied, pulling the depletion region back towards the surface and allowing electrons to accumulate at the heterointerface.

Alternative enhancement mode strategies, including fluorine ion implantation and heterostructure redesign, have been demonstrated in the literature; however, these approaches introduce additional complexity, irreversibility, or reliance on modified epitaxial growth. Gate recession is therefore particularly well suited to this project, as it enables systematic investigation of the relationship between recess depth and threshold voltage using a fixed heterostructure. The development, control and limitations of this approach form the focus of Chapter 5.

CHAPTER 3

Fabrication of Depletion-Mode HEMT Devices

3.1 Overview of the Fabrication Workflow

The depletion-mode HEMTs characterised in this report were fabricated using the standard III-V device process established within the Leeds Nanotechnology Cleanroom. This workflow follows the routine sequence used for GaAs/AlGaAs transistors in this facility: it comprises mesa definition, Ohmic contact formation, contact annealing, and Schottky-gate metallisation. This well-established procedure was followed without modification and carried out over four full-day laboratory sessions. The devices produced through this workflow provide a necessary baseline; they allow for a reliable comparison against which the behaviour of devices with different structures and dimensions can be evaluated later in this project.

3.2 Mesa Definition and Isolation Etch

The fabrication process began with the electrical isolation of individual device regions through mesa etching. Prior to lithography, the GaAs wafer was cleaned sequentially in acetone, isopropanol (IPA), and deionised water (DI) to remove organic residues and ensure sufficient photoresist adhesion. To further enhance the bonding interface between the substrate and the resist, a Surpass primer was spun onto the wafer. This priming step is critical for wet-etching applications: it modifies the surface tension to prevent the etchant from penetrating under the resist edges (an effect known as undercutting) which would otherwise lead to the delamination of the mask. A positive photoresist (S1803) was then spun onto the sample at 3000 rpm, soft-baked on a hot-plate at 115 °C, and subsequently patterned with exposure from the Heidelberg MLA 150 Maskless Aligner (MLA).

Following exposure, the sample was immersed in a caustic Microposit developer solution for 45 seconds, followed by a thorough 2 stage DI water rinse and a nitrogen blow-dry to reveal the mesa patterns. The mesa pattern was then transferred into the wafer using a sulphuric acid, hydrogen peroxide, and water wet-etch solution ($H_2SO_4 : H_2O_2 : H_2O$ in a 1:8:160 ratio). This etchant is highly suited to GaAs/AlGaAs mesa formation: it produces a controllable, moderately fast isotropic etch that removes the active layers uniformly. The rate is sufficiently slow to allow for an accurate timed etch of a shallow ≈ 200 nm recess, yet it remains fast enough to be practical for routine device isolation.

To calibrate the process, an initial 30 s pilot etch was performed; subsequent measurement using an Alpha-Step profilometer revealed an etch depth of 114 nm, corresponding to an initial

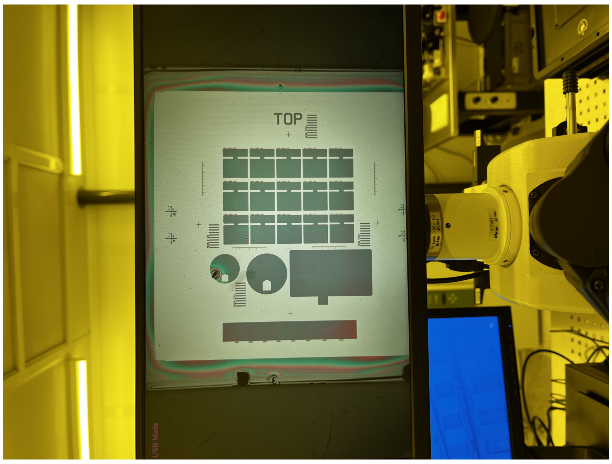

Figure 3.1: Mesa lithography mask used for defining isolated device regions (Post-Development).

etch rate of 3.8 nm s^{-1}. To reach the target depth of 200 nm, a further 86 nm of material required removal. Assuming a constant etch rate, a second etch duration of 22 s was calculated and performed; however, the final measured depth reached only 177 nm. This indicates that the second etch only removed an additional 63 nm at a reduced effective rate of approximately 2.9 nm s^{-1}. While a total depth of 177 nm is sufficient for device isolation (as it successfully penetrates the active GaAs/AlGaAs heterostructure layers), the discrepancy in rates highlights the intrinsic sensitivity of the wet-etch chemistry.

A theory for the observed reduction in the mean etch rate during the second step is attributed to the formation of a native oxide layer on the GaAs surface during the interval in which the sample was removed for profilometry. This oxide layer creates an "incubation period" at the start of the second etch: sample etching is delayed while the etchant penetrates the oxide before reaching the underlying semiconductor. Because the second etch (22 s) was shorter than the first (30 s), this non-productive period represented a larger percentage of the total etch time; consequently, the average rate for that specific step was significantly lowered.

[PROVIDE LITERATURE BACKUP TO THIS THEORY SUGGEST MITIGATION]

3.3 Ohmic Contact Formation

The formation of low-resistance Ohmic contacts is a critical prerequisite for accessing the two-dimensional electron gas (2DEG) within the heterostructure. When metal is deposited directly onto the GaAs cap, the interface naturally forms a Schottky barrier due to Fermi-level pinning at the surface states: this creates a rectifying junction that restricts carrier injection. To overcome this, a specific metallisation scheme and thermal processing step are required to facilitate field-emission (tunnelling) conduction. The objective is to create a heavily doped n^+ region beneath the contact pads that extends down towards the channel; this high doping density significantly narrows the depletion width of the barrier, allowing electrons to tunnel freely between the metal and the semiconductor with linear $I - V$ characteristics.

To define these contacts, a dual-layer photoresist stack comprising Lift-Off-Resist (LOR) beneath S1803 was employed. A single photoresist layer typically develops near-vertical sidewalls:

this causes the evaporated metal to form a continuous film across both the wafer surface and the photoresist edges, making lift-off unreliable and often leaving metallic ”fences” around the contact perimeter. The dual-layer stack mitigates this by exploiting the higher developer solubility of the LOR base layer. During development, the LOR recedes laterally more than the S1803 above it, producing a distinct undercut profile. This physical overhang breaks the continuity of the deposited metal film, ensuring that the metal on top of the photoresist is electrically and mechanically separated from the device contacts. Consequently, solvent-based lift-off yields sharply defined openings free of edge defects.

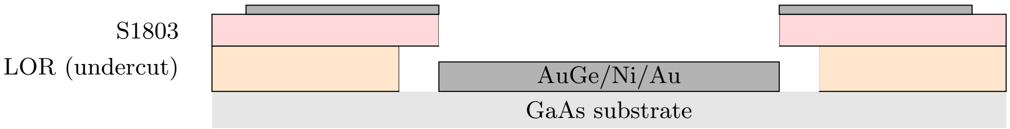

Figure 3.2: Dual-layer LOR/S1803 resist stack after development showing the wider LOR opening that creates an undercut for clean metal lift-off.

Following lithography, the metallisation was performed by the thermal evaporation of a AuGeNi eutectic pellet. The composition of this alloy is functionally specific: Germanium (Ge) serves as the amphoteric dopant which, upon substituting into Gallium sites, acts as a donor; Nickel (Ni) acts as a wetting agent to prevent the Gold-Germanium from ”balling up” during the melt and assists in the diffusion process; and Gold (Au) provides the necessary low-resistance conductivity.

To activate the contacts, the sample underwent Rapid Thermal Annealing (RTA). It is during this thermal cycle that the transition from rectifying to Ohmic behaviour occurs. The heat drives the Germanium atoms to diffuse through the AlGaAs layers and into the GaAs lattice, locally creating the requisite n^+ doping profile. This effectively ”thins” the potential barrier at the interface, establishing a low-resistance tunnelling path that connects the surface metallisation electrically to the buried 2DEG.

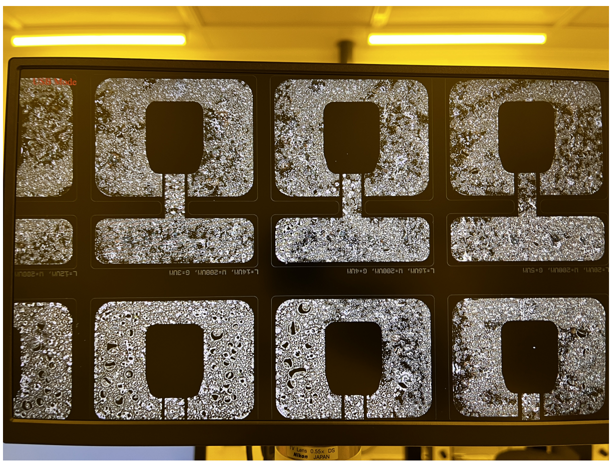

Figure 3.3: Ohmic contacts after lift-off.

3.4 Gate Formation and Final Inspection

With the Ohmic contacts established and annealed, a third photolithography step was performed to define the Schottky gate. As with the previous stage, a dual-layer resist stack was utilised to ensure a clean lift-off. The gate electrode was aligned centrally between the source and drain contacts; its placement is critical, as it provides the electrostatic control required to modulate the carrier density within the channel.

Following development, a Ti/Au stack was evaporated to form the gate electrode. In this configuration, the Titanium (Ti) layer serves as a vital adhesion layer to the semiconductor surface, while the thick Gold (Au) layer provides a stable Schottky barrier and minimises gate resistance. Unlike the Ohmic contacts, the gate metallisation is not annealed; preserving the Schottky barrier height is essential for minimising leakage current and maintaining effective depletion control over the channel.

After metallisation, the resist was removed using solvent lift-off to reveal the final device structure. A small number of lift-off defects (primarily isolated metallic particles) were observed under the microscope and subsequently removed using brief, gentle ultrasonic cleaning. The removal of these artefacts is essential for device reliability: they can otherwise cause electrical short circuits, particularly around the sensitive gate edges.

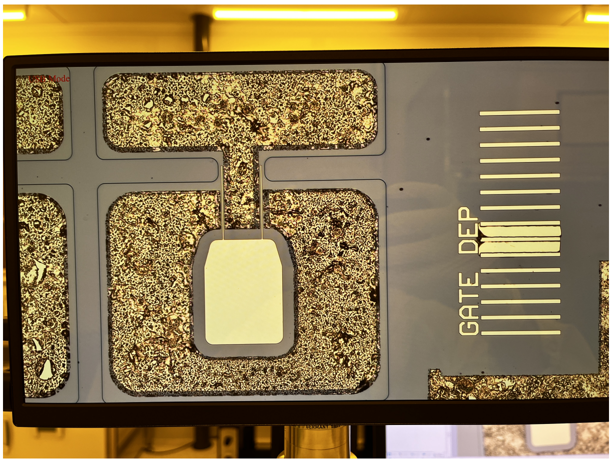

Figure 3.4: Gate contacts after lift-off.

The completed devices were examined using optical microscopy to ensure process integrity. The mesas exhibited clean, well-defined sidewalls; the Ohmic pads were properly positioned; and the gate electrodes showed sharp, uninterrupted edges with no evidence of bridging or residual flakes. Overall, the fabrication process yielded a high-quality array of depletion-mode HEMTs suitable for subsequent electrical characterisation.

CHAPTER 4

Electrical Characterisation and Critical Analysis

CHAPTER 5

Exploration of Enhancement-Mode HEMT Devices

CHAPTER 6

Reflection and Project Management

6.1 Summary of Progress

6.2 Challenges and Adaptations

6.3 Skills Developed

6.4 Lessons for Semester Two

CHAPTER 7

Semester Two Plan

7.1 Gantt Chart

7.2 Planned Technical Deliverables

7.3 Risk Mitigation Strategy

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APPENDIX A

Additional Figures

APPENDIX B

Code and Process Flow